

CSE 0613212: Data Structures and Algorithms I Lab

Part A: Course Information

1. **Program:** B.Sc. in CSE
2. **Course Code:** CSE0613212
3. **Course Title:** Data Structures and Algorithms I Lab
4. **Course Type (GED/Core Course/Electives):** Core Course
5. **Year/Level/Semester/Term:** 2nd year/ 1st Semester
6. **Academic Session:** Autumn 2026
7. **Section:** 3C1
8. **Prerequisite Course (If any):** Structured Programming Language
9. **Credit Value:** 1.50
10. **Contact Hours:** 3.00 hours/Week
11. **Class Schedule:** Sunday (Room: 212, Period: 1:00PM – 4:00PM)
12. **Counseling Schedule:** Tuesday (9:00AM - 12:00PM),
Wednesday (10:30 AM – 12:30 PM)
13. **Course Teacher:** Md. Azharul Karim Chowdhury Anik, Designation: Lecturer, Room: 418(10), Email: azharul.karim@uits.edu.bd , Cell: +8801855234355, Specialization: Machine Learning, Deep Learning, Computer Vision, NLP.
14. **Rationale of the Course:** This laboratory course provides practical experience in implementing and analyzing fundamental data structures and algorithms using C/C++. Students design, implement, test, debug, and evaluate programs involving arrays, linked lists, stacks, queues, searching, sorting, recursion, trees, and graph traversals. Through laboratory exercises, algorithm visualization, debugging activities, and problem-solving tasks, students develop programming proficiency, computational thinking, and the ability to select appropriate data structures and algorithms for solving real-world computational problems efficiently.
15. **Course Objectives:**
 - a. To understand the fundamental concepts and operations of linear and non-linear data structures.
 - b. To analyze the efficiency and behavior of searching, sorting, recursive, and graph traversal algorithms.
 - c. To develop programming skills by implementing data structures and algorithms using structured programming techniques.
 - d. To apply suitable data structures and algorithms for solving computational problems efficiently.

Part B: Skill Mapping

16. Course Learning Outcome (COs)

No.	COs	BT	CP/ WP	CA/ EA	KP/ WK	AT	DM&A
CO1	Demonstrate knowledge of fundamental data structures and algorithms.	cog2	-	-	K2, K3, K4	Q	Discussion
CO2	Implement fundamental data structures and algorithms using programming tools to solve specified computational problems.	cog3	-	-	K6	LP, LT	Demonstration, Discussion, Problem Solving

CO3	Analyze problem requirements and justify the selection of appropriate data structures and algorithms.	cog5	-		K3, K4	P, LT	Discussion
CO4	Design and develop algorithmic solutions for real-world problems using appropriate fundamental data structures and algorithms.	cog6	P1, P2, P3		K5	LT, R, I	Open-ended Problem Solving

*COs- Course Outcomes; BT- Bloom's Taxonomy; Cog- Cognitive Domain; CP- Complex Problems; CA- Complex Activities; AA- Assessment Approach; KP- Knowledge Profile; LT- Lab Test, LF- Lab Final, Q- Quiz, R- Report, P-Presentation, I - Implementation, V- Viva, A- Assignment, CP- Class Performance

17. Mapping of COs with Program Outcomes (POs)

Course Outcomes (COs)	PO (a)	PO(b)	PO(c)	PO(d)	PO (e)	PO(f)	PO(g)	PO(h)	PO(i)	PO(j)	PO(k)	PO(l)
CO1	√											
CO2					√							
CO3		√										
CO4			√									

Part C: Teaching Learning Approach

18. Course plan specifying contents, COs, co-curricular activities (if any), teaching learning and assessment strategy mapped with COs.

Week	Topics	Teaching-Learning Strategy	Assessment Strategy	Corresponding COs
1	Introduction to data structures & Complexity Analysis	Interactive lecture introducing course structure and evaluation policy; demonstration of fundamental data structure concepts using real-life analogies; discussion on time and space complexity basics.	Lab performance	CO1, CO2
2	Array Operations	Interactive lecture on array structure, memory layout, and indexing; live coding of array traversal, insertion, and deletion operations; guided hands-on practice with array-based problem exercises.	Lab performance, Lab test	CO2
3	Matrix Operations	Step-by-step explanation of matrix operations with worked examples; instructor-led coding of matrix arithmetic and transformations; hands-on practice with 2D array implementation	Lab performance, Lab test	CO2
4	Searching	Interactive lecture comparing search	Lab	CO1,

	Algorithms .	strategies using diagrams; comparative implementation of searching algorithms; experimental runtime analysis; guided problem-solving on appropriate algorithm selection. CEP Problem information will be provided	performance , Lab test	CO2
5	Linked List I (Dynamic Memory Allocation + Traversing)	Conceptual explanation of pointers and dynamic memory allocation; live coding of singly linked list creation and traversal; guided hands-on practice with node structures.	Lab performance , Lab test	CO1, CO2
6	Linked List II (Operation on Linked List)	Step-by-step demonstration of insertion and deletion at head, middle, and tail positions; guided coding practice; discussion on pointer manipulation and edge cases; peer code review.	Lab performance , Lab test	CO2
7	Lab Test 1	Conduct of Lab Test 1 to assess understanding of arrays, matrices, searching, and linked lists; brief revision and clarification of common misconceptions based on assessment outcomes.	Lab test	CO2
8	Sorting: Selection Sort, Bubble Sort, Insertion Sort	Interactive lecture and discussion comparing sorting techniques using visual illustrations; instructor-led live coding; guided analysis and comparison of time/space trade-offs across methods to support algorithm-selection discussion.	Lab performance , Lab test	CO1, CO2, CO3, CO4
9	Stack, Applications	Conceptual explanation of stack operations (push/pop) using diagrams; live coding of stack implementation; guided practice converting and evaluating prefix/postfix expressions; open-ended exercise applying stacks to a real-world scenario.	Lab performance , Lab test	CO1, CO2, CO3, CO4
10	Recursion	Interactive explanation of recursive problem-solving using recursion trees; live coding demonstration of Fibonacci and Tower of Hanoi; guided hands-on practice tracing recursive calls; discussion comparing recursive vs. iterative approaches.	Lab performance , Lab test	CO2, CO3, CO4
11	Queues (Linear, Circular, Priority)	Conceptual explanation of queue variants and their real-world use cases; live coding demonstration of each queue type; guided hands-on implementation exercises;	Lab performance , Lab test	CO1, CO2, CO4
12	Graph Representation, Trees, BFS, DFS	Introduction to graph and tree representations using adjacency lists/matrices; live demonstration of BFS and DFS traversal; guided discussion on real-world applications.	Lab performance , Lab test	CO1, CO2, CO3, CO4
13	Report submission and Presentation	Students present and demonstrate their problem implementations; peer review and instructor evaluation; feedback on design	Presentation , Report, Implementat	CO3, CO4

		choices, code quality, and implementation efficiency.-	ion	
14	Lab Test 2, Quiz	Conduct of Lab Test 2 based on week 8-12 and Quiz to evaluate comprehensive understanding of the full course; post-assessment discussion of solutions and strategies.	Lab test, Quiz	CO1, CO3, CO4

Part D: Assessment Approach

19. Assessment and Evaluation

Assessment components		Marks Distribution
Continuous Assessment (100%)	Class Attendance (10) Lab Performance (10)	20%
	Lab Test 1 (10) Lab Test 2 (20)	30%
	Quiz (20) Report (10)	30%
	Implementation (10) Presentation (10)	20%
	Total Marks	

*COs-Course Outcomes; C- Cognitive Domain; P- Psychomotor Domain, A- Affective Domain

20. Assessment Pattern - Continuous Internal Evaluation (100 Marks)

*The percentage distribution of Bloom's categories in the assessment tools may vary by $\pm 5\%$.

Bloom's Category	Attendance & Class Participation (10)	Class Performance (10)	Lab Test-1 (10)	Lab Test-2 (20)	Quiz (20)	Lab Report (10)	Implementation (10)	Project Presentation (10)
Remember								
Understand					20%			
Apply		10%	10%					
Analyze				10%				10%
Evaluate								
Create				10%		10%	10%	

21. Grading system

Numerical Grade	Letter Grade	Grade Point
80% and above	A + (A plus)	4.00
75% to less than 80%	A (A regular)	3.75
70% to less than 75%	A – (A minus)	3.50
65% to less than 70%	B + (B plus)	3.25
60% to less than 65%	B (B regular)	3.00
55% to less than 60%	B – (B minus)	2.75
50% to less than 55%	C + (C plus)	2.50
45% to less than 50%	C (C regular)	2.25
40% to less than 45%	D	2.00

Less than 40%	F	0.00
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Part E: References

22. Learning Materials

- 1) Recommended Readings
 - a) M. A. Weiss, *Data Structures and Algorithm Analysis in C++*, 4th ed. Boston, MA, USA: Pearson, 2014.
- 2) Supplementary Readings
 - a) D. E. Knuth, *The Art of Computer Programming*, vols. 1–4A, 3rd ed. Boston, MA, USA: Addison-Wesley, 1997–2011.

Part F: Additional Information

1. Program Outcome: <https://uits.edu.bd/departments-of-computer-science-engineering/>
2. Outcome Based Curriculum: <https://uits.edu.bd/curriculum-of-cse/>
3. Course Catalog: <https://uits.edu.bd/course-catalog-of-computer-science-engineering/>
4. Faculty Information: <https://uits.edu.bd/faculty-members-of-cse/>
5. Assessments Rubrics:
6. Details of Bloom Taxonomy, Knowledge Profile, Complex Engineering Problems and Complex Engineering Activities:
7. Make-up Exam Procedure:
8. Attendance Policy:
9. Academic Calendar: <https://uits.edu.bd/academic-calendar/>
10. Academic Policies:
11. Proctorial Rules (Code of Conduct): <https://uits.edu.bd/wp-content/uploads/2024/01/Code-of-Conduct.pdf>